

Decentralized Digital Identity and Government-ID Sharing for Healthcare

Project report, Introduction to Blockchains

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1. Introduction

WRITER D, in your own words (prompts: `report-draft.md` §1, about 1 page): the problem in our domain, how existing systems handle it (cite [1]–[10]), our angle, aims and objectives, scope and assumptions. Don't give the solution away here.

2. Architecture

2.1 Actors and permissions

WRITER D + A (prompts: `report-draft.md` §2.1).

Table 1: Roles and permissions

Role	Initiates	Allowed	Not allowed
Identity owner (patient)	Registration, document updates, consent grants and revocations	Register hashes + gatekeeper link; grant scoped consent (1–365 days) to whitelisted requesters; revoke at any time; read the access log; earn 10 ACT on the first grant to each requester	Change anyone else's record or consent
Requester (provider)	Access requests	Once whitelisted: call <code>requestAccess</code> ; with its own fresh GRANTED transaction, fetch the files the scope allows from the gatekeeper	Give itself consent; get files without valid consent; alter or suppress log entries; gain access through ACT
Administrator	Deployment, whitelist, reward rate	Deploy and wire the contracts once; whitelist or remove requesters; set the reward amount	Get files without consent; grant or revoke consent; write or delete log entries; mint ACT; re-wire minter or logger

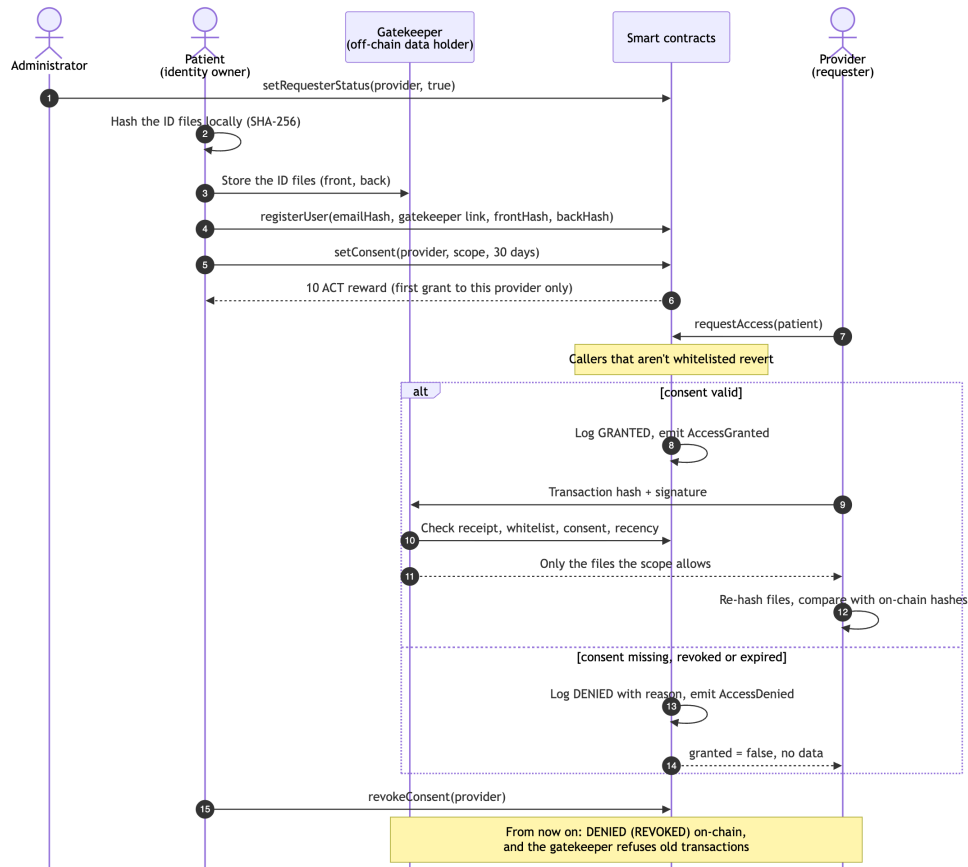


Figure 1: Interaction overview: registration, consent, access, revocation.

2.2 Functional requirements

WRITER D (prompts: report-draft.md §2.2; source: Step 1 §1.4, 12 requirements).

2.3 Data model

WRITER D (prompts: report-draft.md §2.3).

Table 2: Data model: what is stored where

Attribute	On-chain?	Off-chain?	Hashed?
Wallet address	Yes	—	No (already a pseudonymous identifier)
Full name, date of birth, ID number	No	Yes (inside the ID files)	No
Email	No	Yes (local profile)	SHA-256 on-chain, for the uniqueness check
Document link (gatekeeper URL)	Yes	—	No
Front / back of the ID	No	Yes (local data store, released only by the gatekeeper)	SHA-256 of each file on-chain
Consent record (requester, scope, grant/expiry time, revoked)	Yes	—	No (contracts must read it)
Access log entry (requester, time, outcome, reason)	Yes	—	No
ACT balances	Yes	—	No

2.4 Consent model

WRITER D (prompts: report-draft.md §2.4).

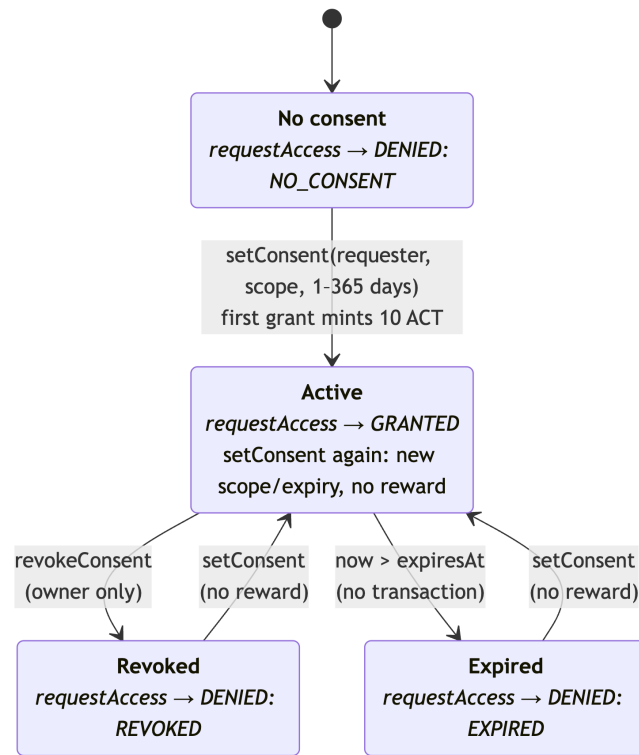


Figure 2: Consent lifecycle. Only the first grant to a requester mints ACT; expiry needs no transaction.

2.5 Audit log

WRITER D (prompts: report-draft.md §2.5).

2.6 System architecture

WRITER A + C (prompts: report-draft.md §2.6).

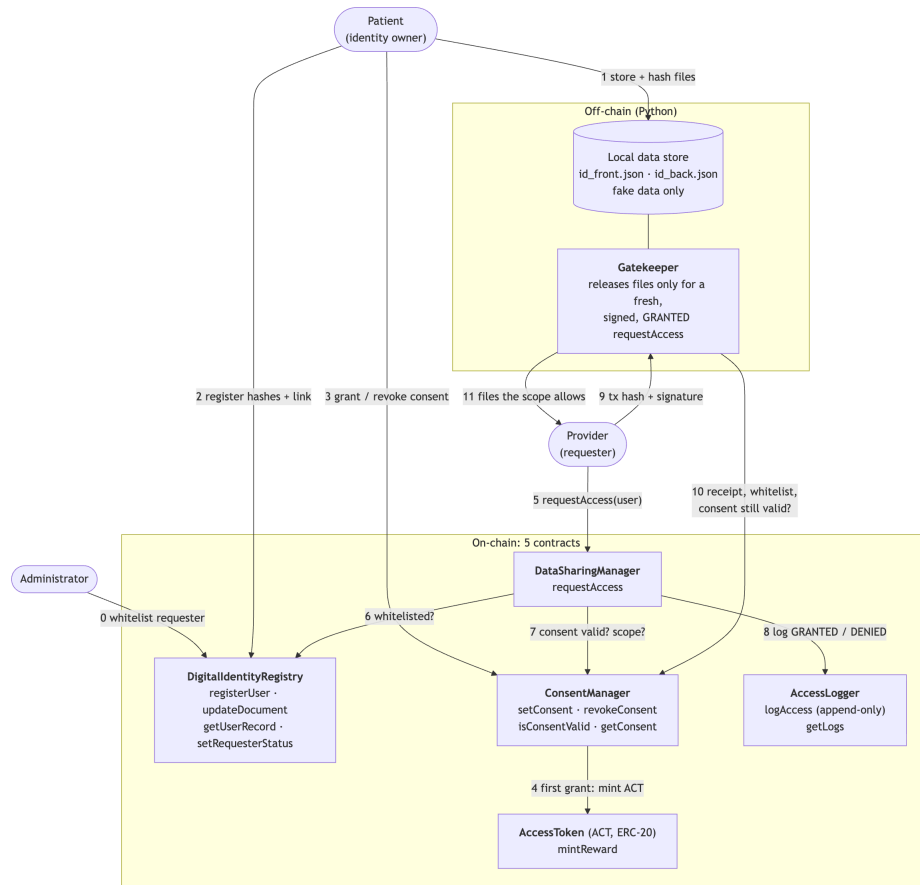


Figure 3: Architecture: five contracts, the local data store and the gatekeeper (steps 0–11).

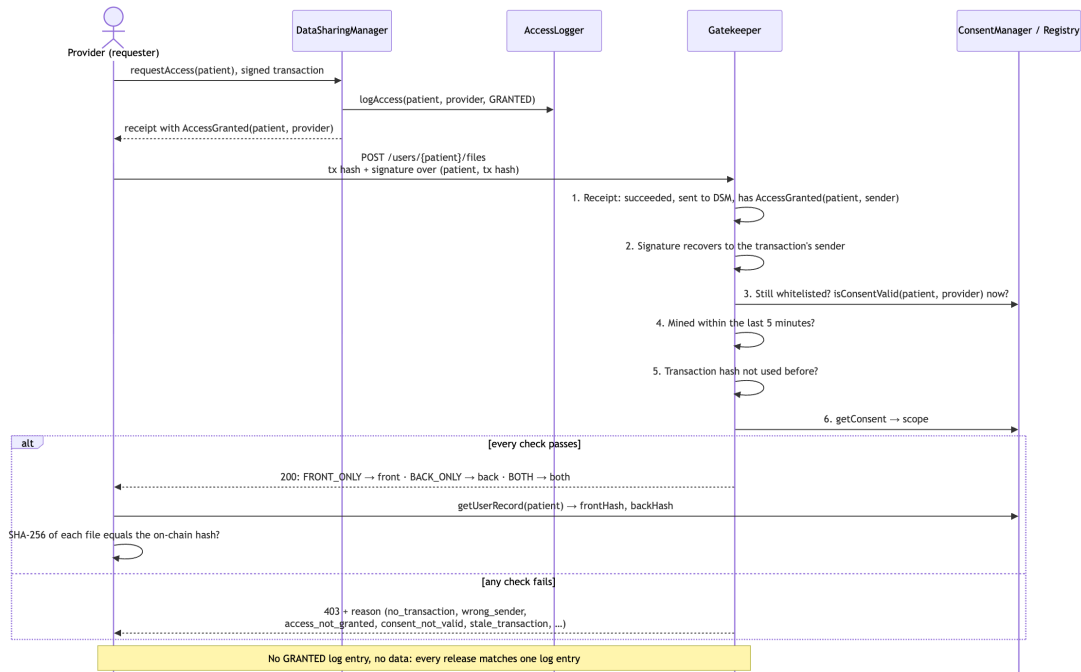


Figure 4: Data access through the gatekeeper: no GRANTED log entry, no data.

3. Implementation

3.1 Tooling

WRITER A (prompts: report-draft.md §3.1).

3.2 Smart contracts

WRITER A (prompts: report-draft.md §3.2: the design decisions and their reasons).

3.3 Off-chain components (Python)

WRITER C (prompts: report-draft.md §3.3).

3.4 Deployment

WRITER A (prompts: report-draft.md §3.4). Point to the README for the steps.

4. Experimental Results

4.1 Testing approach

WRITER B (prompts: report-draft.md §4.1).

4.2 Test results

WRITER B: interpret the table (80/80 Solidity and 15/15 gatekeeper tests pass; 100 % line coverage of the five contracts). The full per-test tables are in the appendix.

Table 3: Tested functionality and why it is critical. Test numbers as in Appendix A; R Registry, C ConsentManager, T Access-Token, L AccessLogger, D DataSharingManager, I integration, G gatekeeper

Functionality	Tests	Why it is critical
Registration stores only hashes + link; empty values rejected	R2–4, R7–9	The chain is public: registration must never store personal data, and an empty hash couldn't be checked later
One identity per address and per email hash	R5, R6	Prevents duplicate or impersonating identities
Document updates by the owner only	R10–13, I76	A renewed ID replaces the old hashes; only the owner may change it
Admin-only requester whitelist	R1, R14, R15	Only vetted providers may receive consent or request access
Consent is scoped, 1–365 days, per requester	C16–19, C23, C24	Users decide who sees what and for how long
Consent only for registered users and whitelisted requesters	C20–22	No consent to typo'd, malicious or removed addresses
Expiry happens by itself at expiresAt	C25–27, I75	Access stops on time without a cleanup transaction
Only the owner revokes, once, immediately	C28–33	Revocation is the user's main control
ACT reward once per (user, requester)	C34–39	Incentive without token farming
Only ConsentManager mints; one-time wiring	T40–48, I79	The admin can't mint ACT

Functionality	Tests	Why it is critical
Only DataSharingManager logs; one-time wiring	L49–53, I79	Nobody can forge log entries
Log is append-only, ordered, per user, public	L54–59	An audit trail is only trustworthy if entries can't change
Granted access returns exactly the scope, and is logged	D60–64	Scope is enforced; every access leaves a trace
Each denial is logged without reverting	D65–69, I74, I75	A revert would erase the DENIED entry
Tokens never grant access	D70	Access comes from consent only
Non-whitelisted callers revert; removed requesters lose access	D72, D73, I78	No log spam; untrusted providers are cut off
Full workflows across all contracts	I74–80	The contracts work together as designed, users stay isolated
Gatekeeper releases files only for a fresh, signed GRANTED transaction	G1–G15	The public link alone protects nothing; the gatekeeper does

4.3 Gas costs and optimisation

Table 4: Deployment cost per contract (gasUsed of the deployment transaction)

Contract	Gas before	Gas after	Change	Bytecode (bytes)
AccessToken	696,804	696,804	0.0 %	2,615
DigitalIdentityRegistry	772,744	759,768	−1.7 %	3,148
ConsentManager	702,661	729,611	+3.8 %	2,897
AccessLogger	634,505	514,920	−18.8 %	2,016
DataSharingManager	613,162	507,916	−17.2 %	2,093
Total	3,419,876	3,209,019	−6.2 %	

Table 5: Average gas per function before and after optimisation (gasUsed of real transactions, 5 users)

Function	Before	After	Change	What changed
registerUser	228,236	205,958	−9.8 %	registered derived from the email hash: one slot fewer
setConsent (first grant)	162,670	96,681	−40.6 %	Consent record packed into 1 slot instead of 4
setConsent (re-grant)	48,354	39,265	−18.8 %	1 slot rewritten instead of 3
revokeConsent	28,811	28,825	0.0 %	Not targeted
requestAccess (granted)	145,912	89,230	−38.8 %	Enum reason, one consent read, packed log entry and consent
requestAccess (denied)	127,968	72,302	−43.5 %	Same
setRequesterStatus	47,779	47,779	0.0 %	Not targeted

WRITER B (prompts: report-draft.md §4.3): which operations are expensive and why, which optimisation helped most (packing), and the events-only trade-off (61k / 44k gas per access, but no on-chain reads; see step4-deliverable.md §2.5).

4.4 Simulation and scalability

Table 6: Scaling in cost and time on the local Hardhat node (M = 3 requesters)

Users (N)	Total txs	Avg gas / tx	Total gas	Avg confirmation, instant mining	Total time, instant mining	Avg confirmation, 12 s blocks	Total time, 12 s blocks
5	28	96,206	2,693,763	2.9 ms	0.63 s	11.94 s	72.3 s
10	53	98,300	5,209,898	3.4 ms	1.19 s	11.89 s	72.5 s
25	128	100,076	12,809,771	2.6 ms	2.78 s	11.65 s	73.2 s
50	253	100,696	25,476,186	2.7 ms	5.65 s	11.48 s	74.3 s

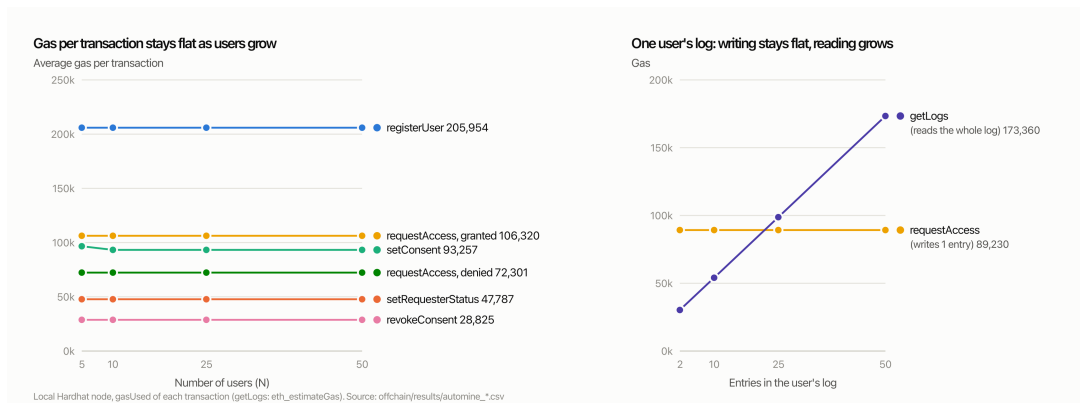


Figure 5: Gas per operation stays flat as the number of users grows; reading one user's whole log grows with its length.

WRITER C (prompts: report-draft.md §4.4): the setup, whether cost per operation stays flat (it does), what grows (total gas and time linearly in N ; getLogs read cost with the length of one user's log), and what local timing does and doesn't say about a real network (Lecture 6).

5. Discussion and Conclusion

5.1 What works

WRITER D, with input from everyone (prompts: report-draft.md §5.1).

5.2 Limitations

WRITER D, with input from everyone (prompts: report-draft.md §5.2, including the A3 decision about the public log).

5.3 Next steps

WRITER D (prompts: report-draft.md §5.3).

5.4 Conclusion

WRITER D (prompts: report-draft.md §5.4): 4–6 sentences, nothing new.

6. Contributions and AI use

EVERYONE writes their own line: name, workstream, what they built or wrote. Then the AI transparency statement in the coursebook template (prompts and the list of AI-assisted parts: report-draft.md §6).

References

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Web sources accessed on 1 October 2026. Keep only the entries the text cites.

Appendix A. All tests

The full per-test tables (80 Solidity tests and 15 gatekeeper tests, each with what it checks and why it is critical) are in `docs/step4-deliverable.md` §1.3–1.4. Paste them here if the appendix doesn't count towards the page limit (ask a TA), otherwise point to the code archive.